Player Community, Moderation & Engagement Platform

1. Preface

This document specifies the software requirements for the Player Community, Moderation & Engagement Platform.
It is intended for system users, developers, testers, and evaluators to understand what the system should do.

2. Introduction

2.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the Player Community, Moderation & Engagement Platform, based on stakeholder needs and system modeling using use-case and activity diagrams.

2.2 Scope

The platform supports gaming communities by enabling:

- Community and channel creation
- Real-time communication and interaction
- Content reporting and moderation
- Role-based administration and governance
- Engagement through events and notifications

Moderation automation using system bots is treated as a supporting subsystem under the same domain.

2.3 Definitions

Term	Description
Platform	Player Community, Moderation & Engagement Platform
Community	A group formed around a game or shared interest for interaction and engagement
Channel	A dedicated communication space within a community for discussions or announcements
Role	A predefined set of permissions and responsibilities assigned to a user
Moderation	The process of monitoring user activities and enforcing platform or community rules
Moderation Action	An action taken to enforce rules, such as warnings, content removal, or restrictions
Report	A request raised to flag inappropriate behavior or content for review

3. Overall Description

3.1 System Users

- Audience
- Gamer
- Community Manager
- System Moderator
- System Administrator

3.2 Product Perspective

The Player Community, Moderation & Engagement Platform is a centralized community management system that integrates:

- Community management
- Communication and interaction
- Moderation and reporting
- Engagement tracking
- Notifications and logging

It is not responsible for game engine logic or gameplay control.

4. User Requirements (High-Level)

UR1
The system shall allow the Audience to view community activities, attend events, and report inappropriate content.

UR2
The system shall allow the Gamer to create or join communities and channels and communicate with other users.

UR3
The system shall allow the Community Manager to manage communities, assign roles, and host events.

UR4
The system shall allow the System Moderator to review reported content and enforce platform-wide rules.

UR5
The system shall allow the System Administrator to manage platform settings, policies, and user roles.

Software Requirements Specification

1. Functional Requirements

FR-1 User Authentication & Role Management

- The system shall allow users to log in as Gamer, Audience, Community Manager, System Moderator, or System Administrator.
- The system shall authorize actions based on the user's role.
- Unauthorized users shall be denied access to restricted features.

FR-2: Community and Channel Management

- The system shall allow a Gamer to create and publish a game community after validating community details.
- The system shall assign the creator as the Community Owner and make published communities visible to other users.
- The system shall allow the Community Owner to create multiple channels within a community.
- The system shall allow configuration of channel types and permissions.
- The system shall store and manage community and channel details in the database.

FR-3 Discover and Join Community

- The system shall allow Audience users to browse available communities.

- The system shall display community details and rules.
- The system shall allow Audience users to join a community.
- The system shall store community membership information.

FR-4 Channel Interaction and Engagement

- The system shall allow users to post messages and react to content in channels.
- The system shall update the activity feed in real time.
- The system shall award XP for valid user interactions.
- The system shall store messages and reactions in the database.

FR-5 Event Monitoring and Rule Enforcement

- The system shall allow the Community Manager to monitor live events.
- The system shall detect rule violations during events.
- The system shall allow warning, muting, or penalizing participants.
- The system shall update participant status in the database.

FR-6 Reporting User or Content

- The system shall allow Gamer and Audience users to report content or users.
- The system shall allow selecting report reasons and adding descriptions.
- The system shall generate a unique report ID.
- The system shall store reports in the database.
- The system shall notify the System Moderator.

FR-7 Report Review and Action

- The system shall allow the System Moderator to review reports.
- The system shall allow validating or closing reports.
- The system shall allow actions such as warn, mute, or temporary ban.
- The system shall notify the reporting user and the reported user.
- The system shall update reports and user status in the database.

FR-8 Feedback and Analytics

- The system shall collect feedback after events.
- The system shall store feedback in the database.
- The system shall allow the System Administrator to view platform metrics.
- The system shall aggregate analytics from stored data.

FR-9 Platform Administration

- The system shall allow the System Administrator to configure moderation rules.
- The system shall allow updating automated moderation settings.
- The system shall maintain audit logs for moderation actions.

2. Non-Functional Requirements

NFR-1 Performance

- The system shall respond to user actions within acceptable time limits.
- The system shall handle multiple users interacting simultaneously.

NFR-2 Scalability

- The system shall support growth in users, communities, channels, and events.
- The system architecture shall allow future feature expansion.

NFR-3 Security

- The system shall enforce role-based access control.
- The system shall protect user data and reports from unauthorized access.
- The system shall securely store authentication and activity data.

NFR-4 Reliability

- The system shall ensure data consistency across all operations.
- The system shall prevent data loss during failures.
- The system shall maintain logs for recovery and auditing.

NFR-5 Usability

- The system shall provide an intuitive and user-friendly interface.
- The system shall clearly guide users through community, channel, and event workflows.

NFR-6 Maintainability

- The system shall follow MVC architecture.
- The system shall use modular and readable code.
- The system shall allow easy debugging and updates.

NFR-7 Compatibility

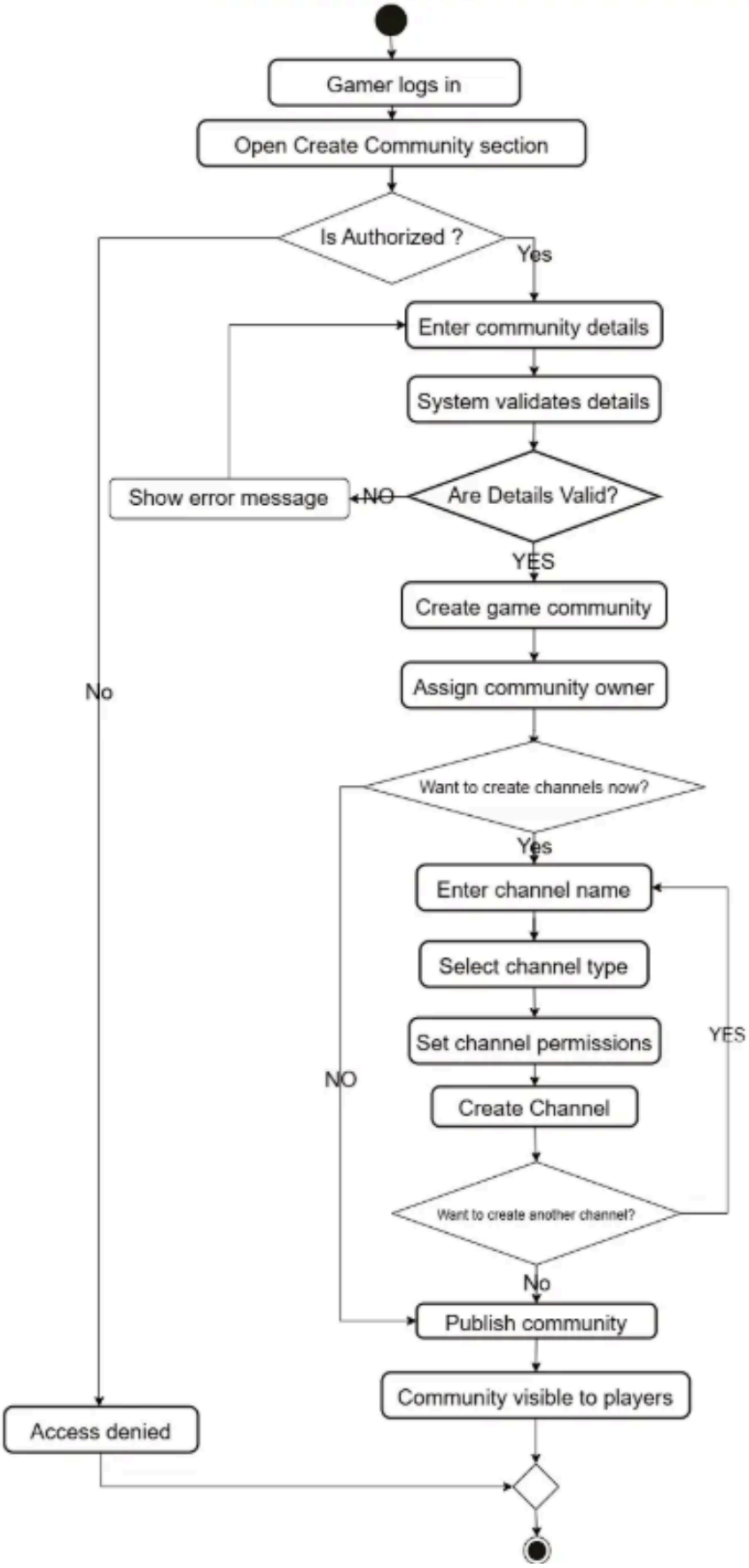
- The system shall run on modern web browsers.
- The system shall be accessible on different screen sizes.

UML Diagrams

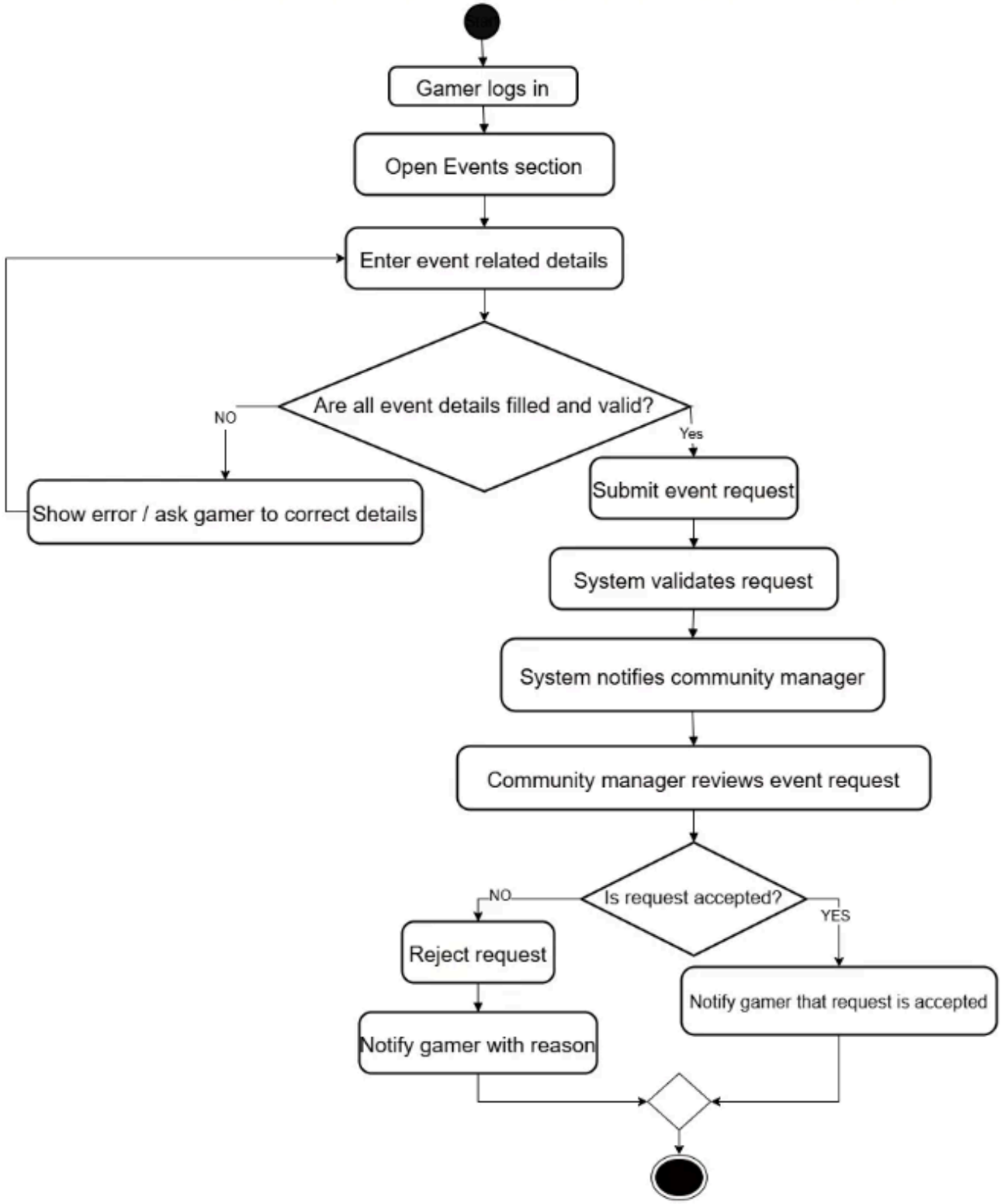
1.Use Case Diagram

2. Activity Diagrams

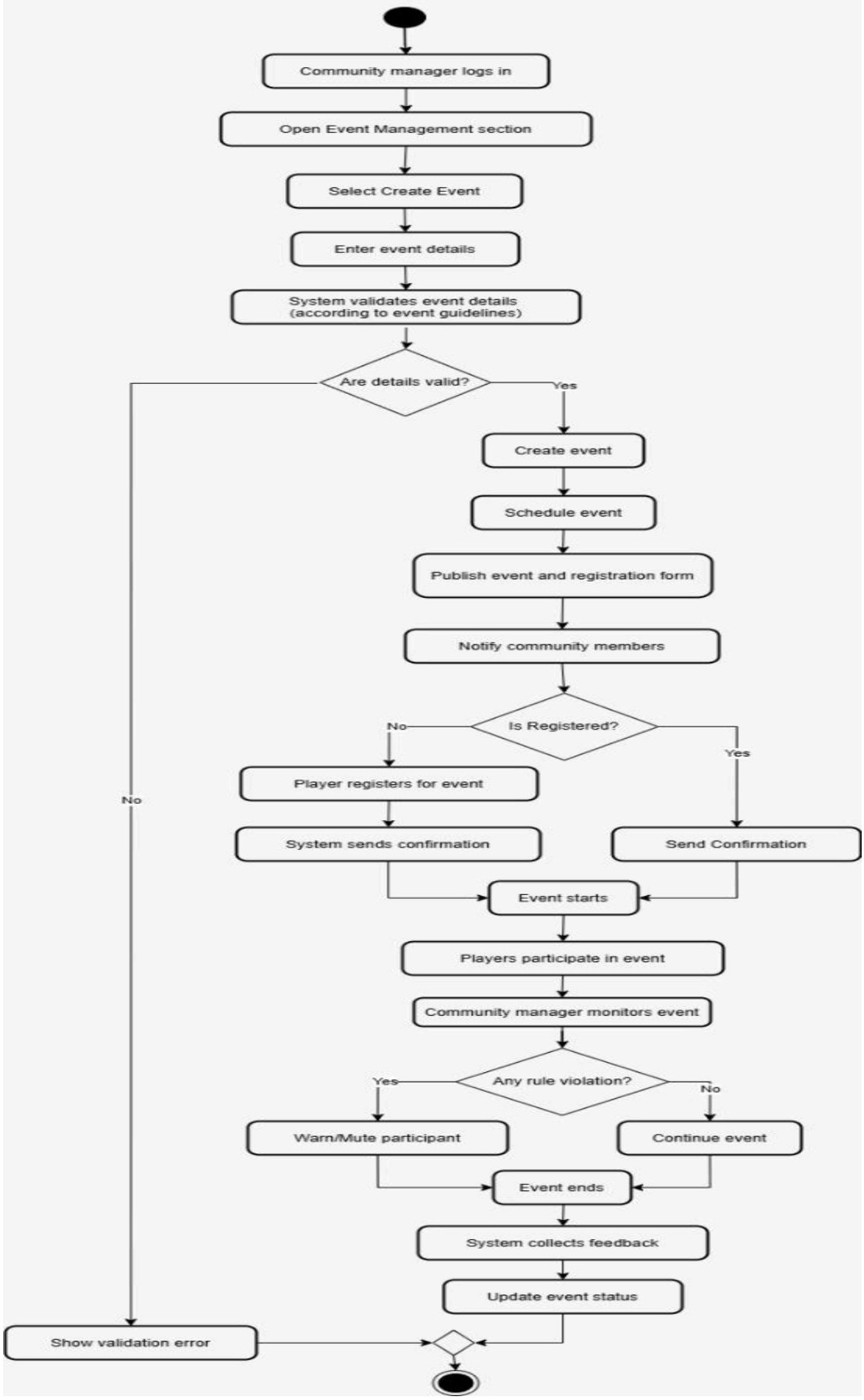
Activity Diagram: Create Game Community & Channel



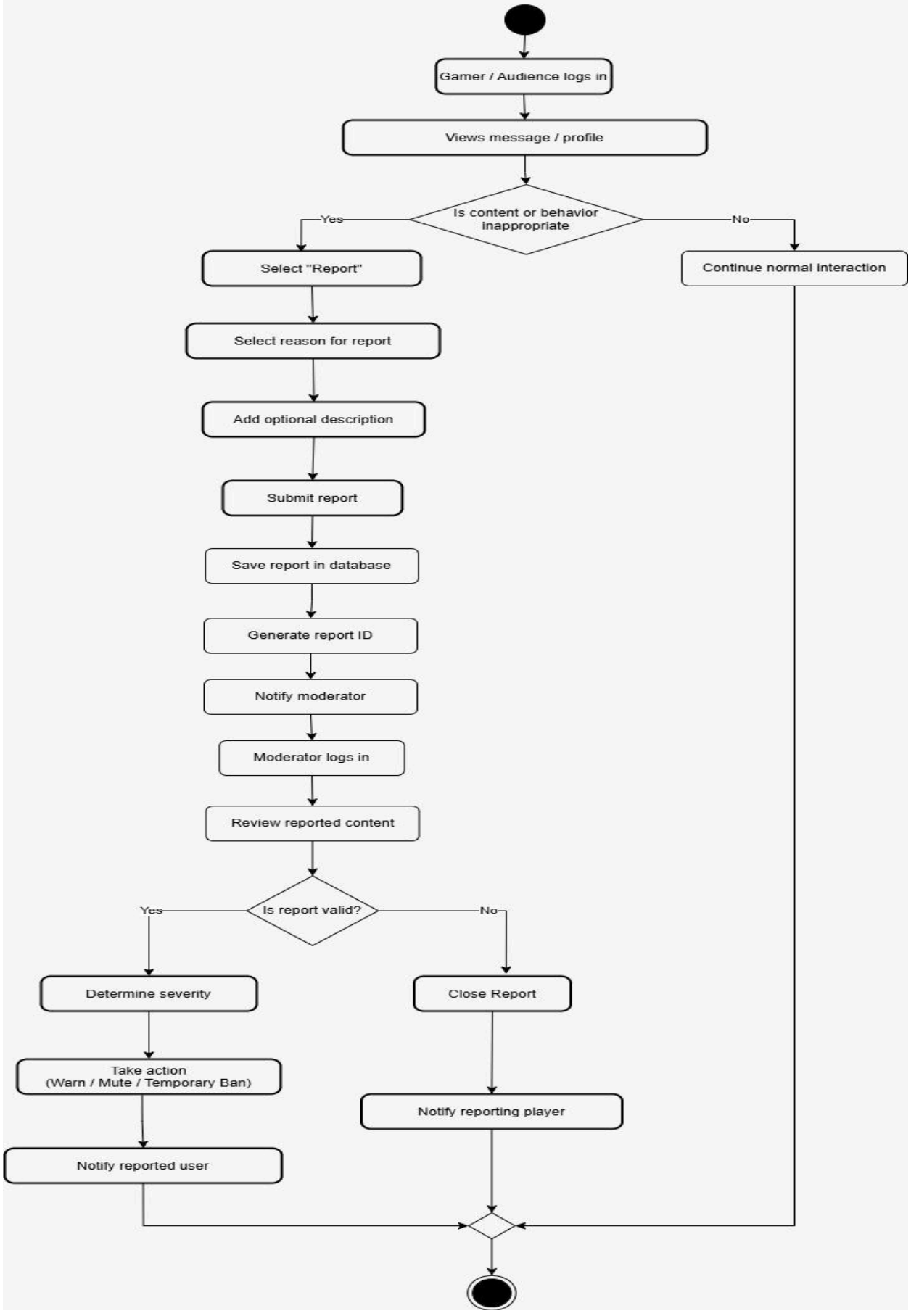
Activity Diagram : Event Request & Approval



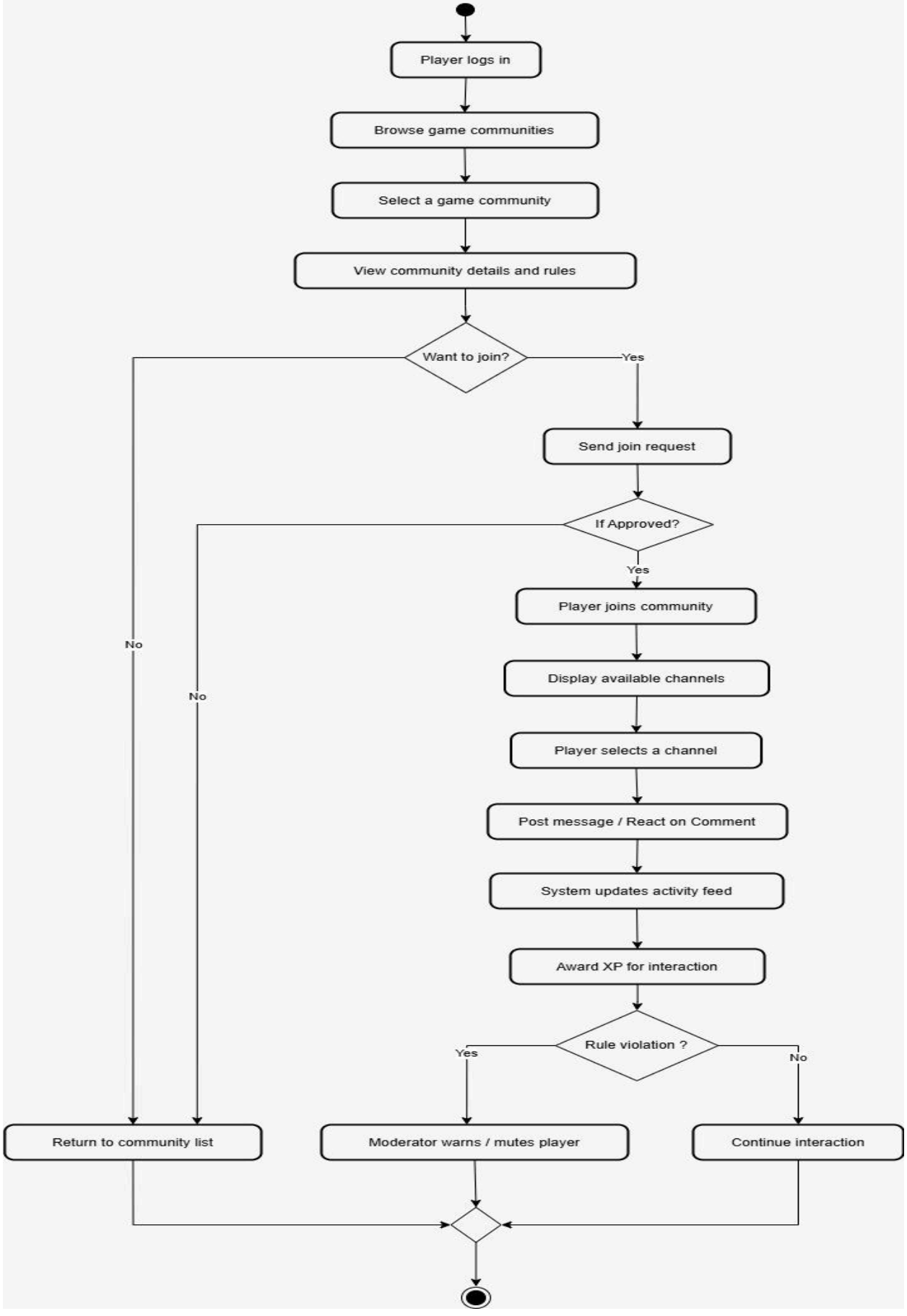
Activity Diagram: Create Community Event and member participation



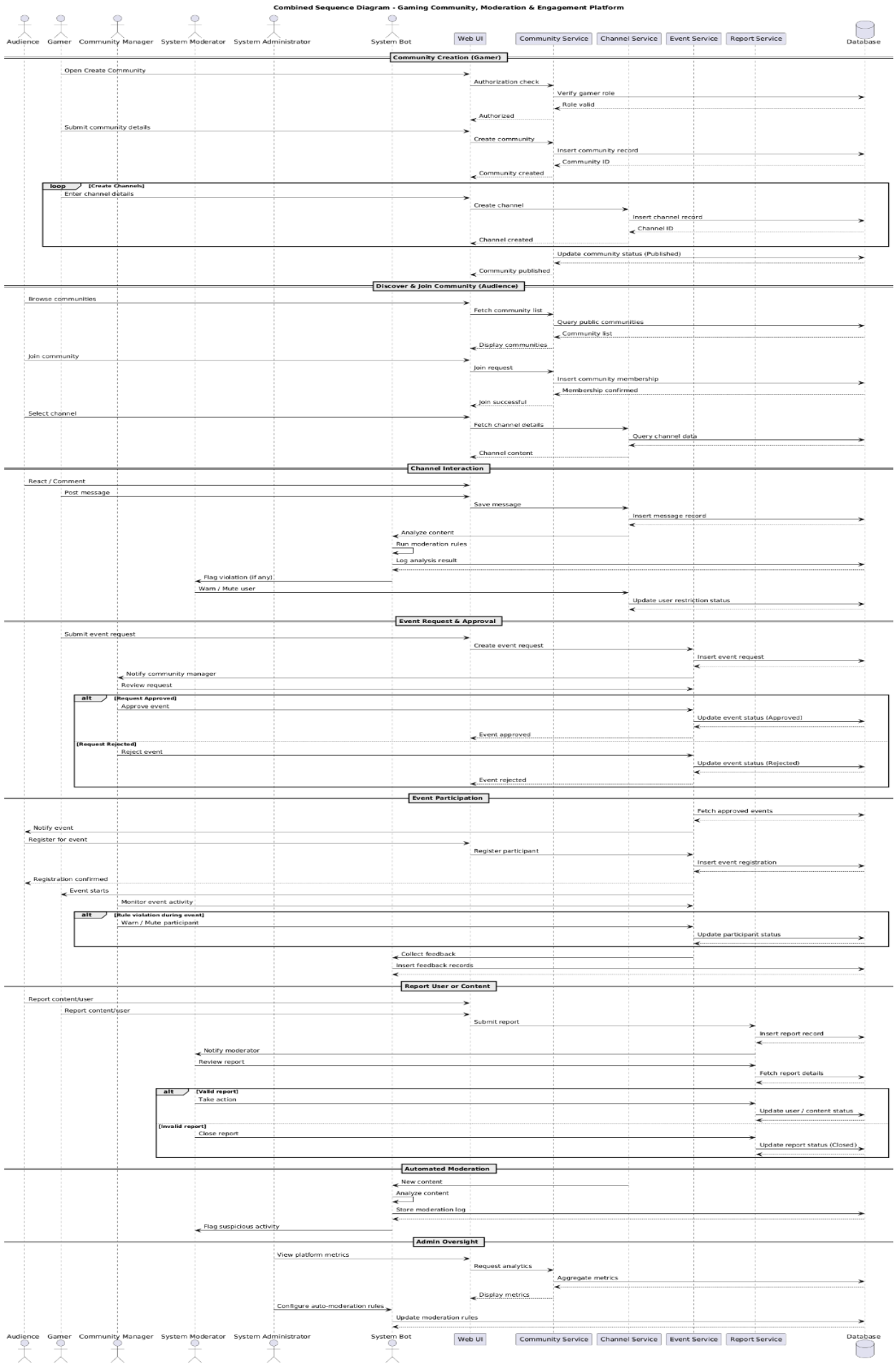
Activity Diagram - Report User or Content and taking action



Activity Diagram: Join Game Community & Interact



3. Sequence Diagram


Combined Sequence Diagram - Gaming Community, Moderation & Engagement Platform
Audience
Gamer
Community Manager
System Moderator
System Administrator
System Bot
Web UI
Community Service
Channel Service
Event Service
Report Service
Database
Community Creation (Gamer)
Open Create Community
Authorization check
Verify gamer role
Role valid
Authorized
Submit community details
Create community
Insert community record
Community ID
Community created
loop [Create Channels]
Enter channel details
Create channel
Insert channel record
Channel ID
Channel created
Update community status (Published)
Community published
Discover & Join Community (Audience)
Browse communities
Fetch community list
Query public communities
Community list
Display communities
Join community
Join request
Insert community membership
Membership confirmed
Join successful
Select channel
Fetch channel details
Query channel data
Channel content
Channel Interaction
React / Comment
Post message
Save message
Insert message record
Analyze content
Run moderation rules
Log analysis result
Flag violation (if any)
Warn / Mute user
Update user restriction status
Event Request & Approval
Submit event request
Create event request
Insert event request
Notify community manager
Review request
alt [Request Approved]
Approve event
Update event status (Approved)
Event approved
[Request Rejected]
Reject event
Update event status (Rejected)
Event rejected
Event Participation
Fetch approved events
Notify event
Register for event
Register participant
Insert event registration
Registration confirmed
Event starts
Monitor event activity
alt [Rule violation during event]
Warn / Mute participant
Update participant status
Collect feedback
Insert feedback records
Report User or Content
Report content/user
Report content/user
Submit report
Insert report record
Notify moderator
Review report
Fetch report details
alt [Valid report]
Take action
Update user / content status
[Invalid report]
Close report
Update report status (Closed)
Automated Moderation
New content
Analyze content
Store moderation log
Flag suspicious activity
Admin Oversight
View platform metrics
Request analytics
Aggregate metrics
Display metrics
Configure auto-moderation rules
Update moderation rules